

```
output += "Square of " + x +  
" is " + result + "\n";  
}  
outputArea.setText( output );  
}  
public int square( int y )  
{  
    return y * y;  
}  
}
```

Notice this syntax: We are inside method `init()`, yet we're calling to another method `square()` without referring to an object. "Methods in a class definition are allowed to invoke all other methods in the same class this way." Inherited methods can be called this same way.

All variables declared inside a method are local variables. And local variables must be initialized. They live only while the method is being executed and vanish after the method is done.

```
public double sum( double x, double y )  
{  
    double sum = 0;  
    sum = x + y;  
    return sum;  
}
```

In this example, all three variables are local. The variables `x` and `y` declared as parameters in the header, and the variable `sum` declared in the body of the method.

3.1 Coercion of Arguments

```
public double sum( double x, double y )  
{  
    double sum = 0;  
    sum = x + y;  
    return sum;  
}
```

What would happen if--when we called this method--we passed it integers, rather than the doubles it expects? If it was able to, the compiler would attempt to convert or "coerce" the arguments into the right type. If I passed